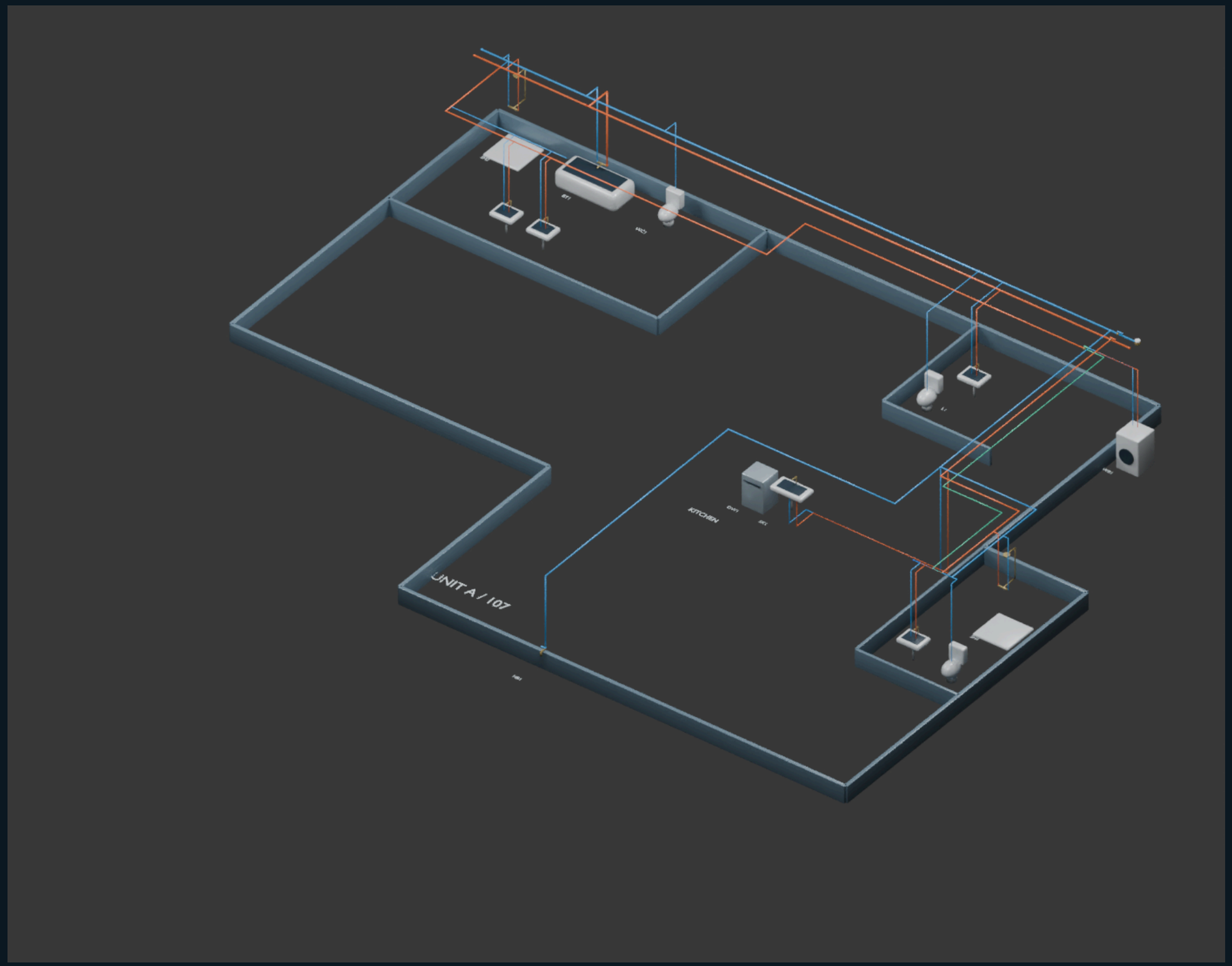
HIGHLAND / DOMESTIC WATER

Unit 107: scale tied to the construction plans

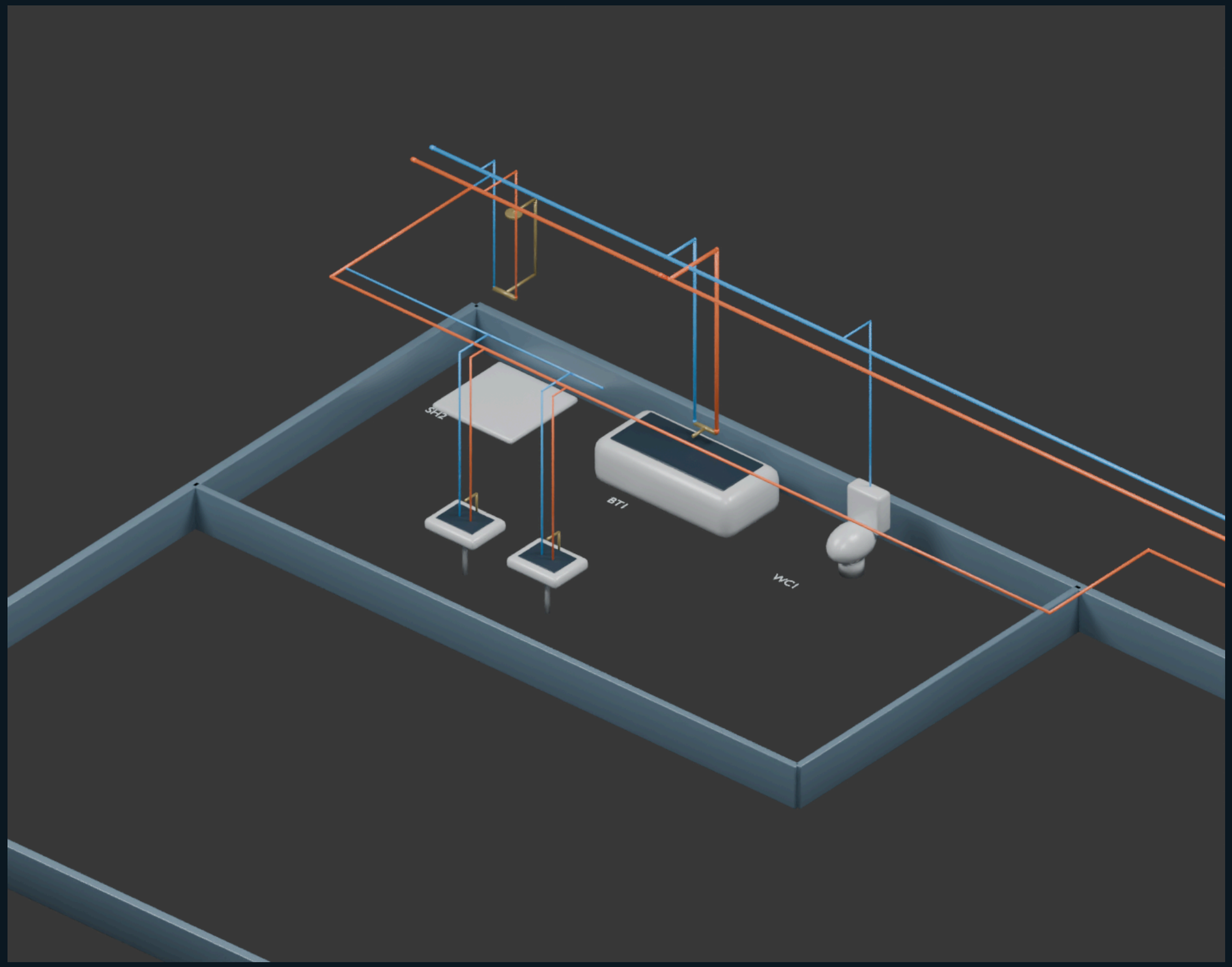
Unit XY is now provisionally calibrated through A2.03 structural grid dimensions and the P2.02 plumbing plan. Colors: blue cold, orange hot, green circulation return. Pipe heights and fixture envelopes remain assumed.



MODEL STUDY | Unit XY provisionally calibrated; pipe heights and envelopes assumed. Not an as-built.

Primary bathroom: supply connections

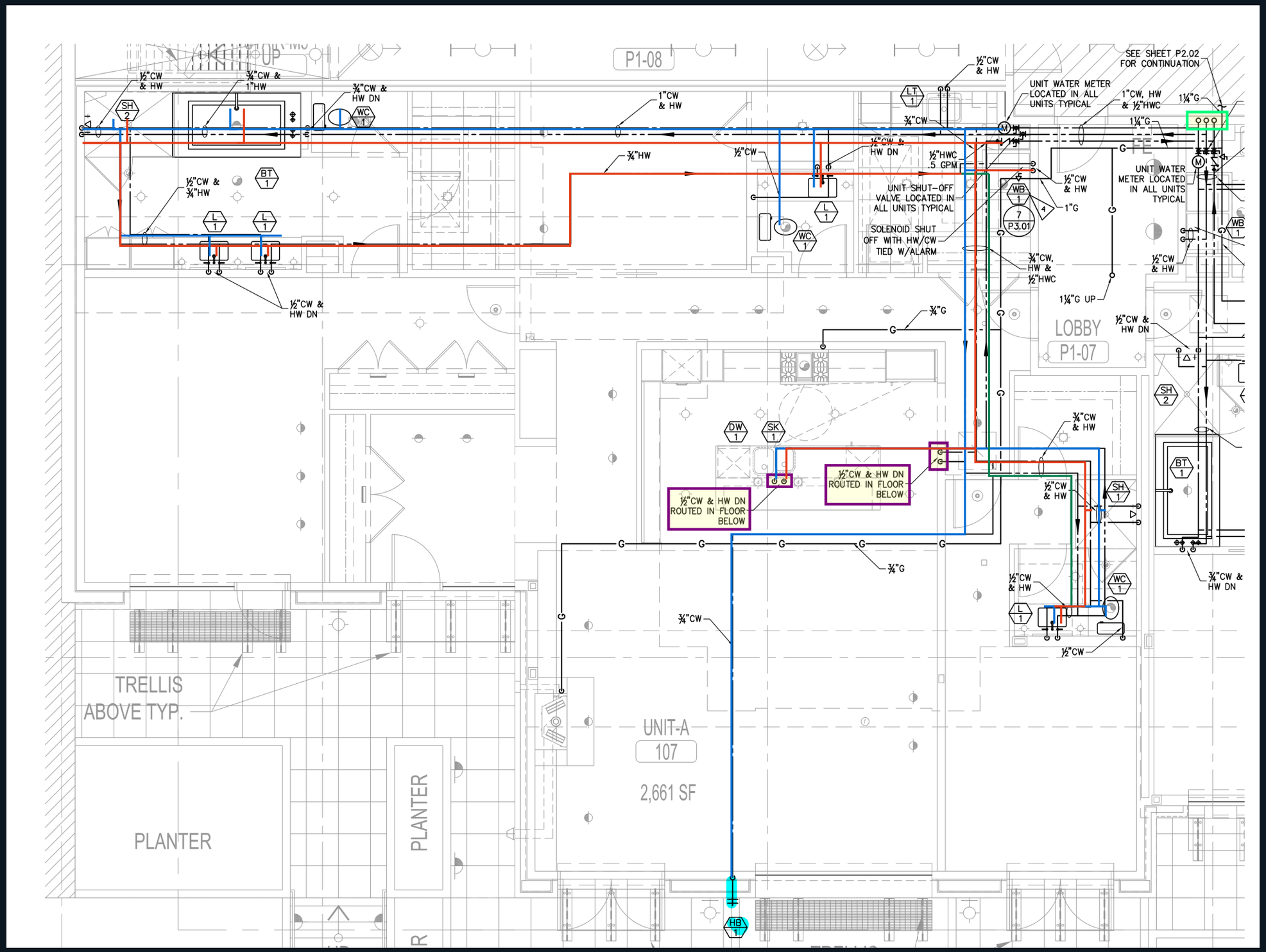
The calibrated plan trace places the two lavatories, bathtub, shower and toilet in relation to their supplies. The 11 ft floor-to-floor height is documented; clear ceiling height, pipe heights and fixture body dimensions are not.



MODEL STUDY | Unit XY provisionally calibrated; pipe heights and envelopes assumed. Not an as-built.

Check the route before extending the model

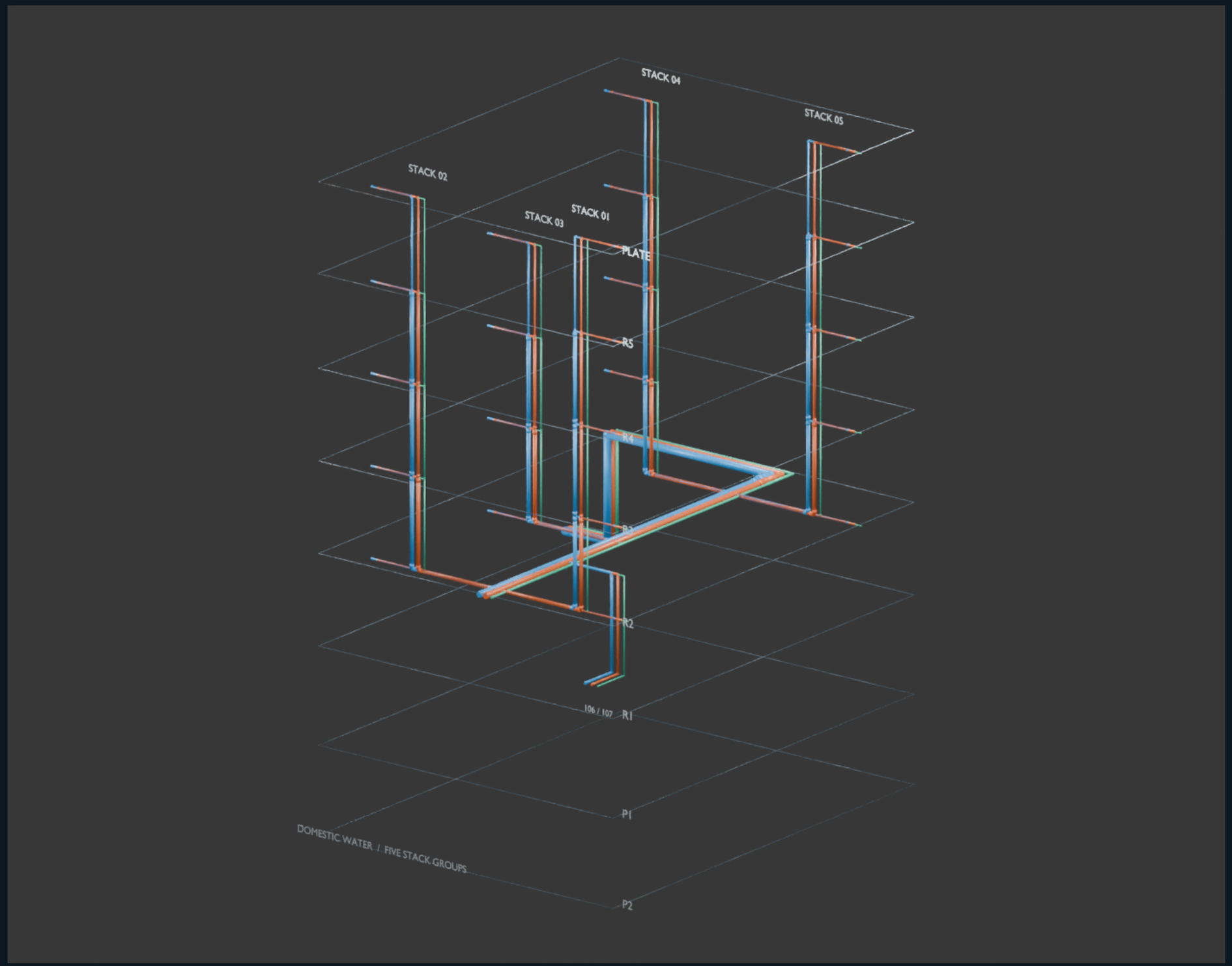
Overlay generated from the calibrated specification and projected back to P4.01. Scale allowance: +/-1%; simplified local trace allowance: +/-6 in. Kitchen underfloor path and exact loop joins remain assumptions. Not a field survey.



MODEL STUDY | Unit XY provisionally calibrated; pipe heights and envelopes assumed. Not an as-built.

The shared network: documented floor datums

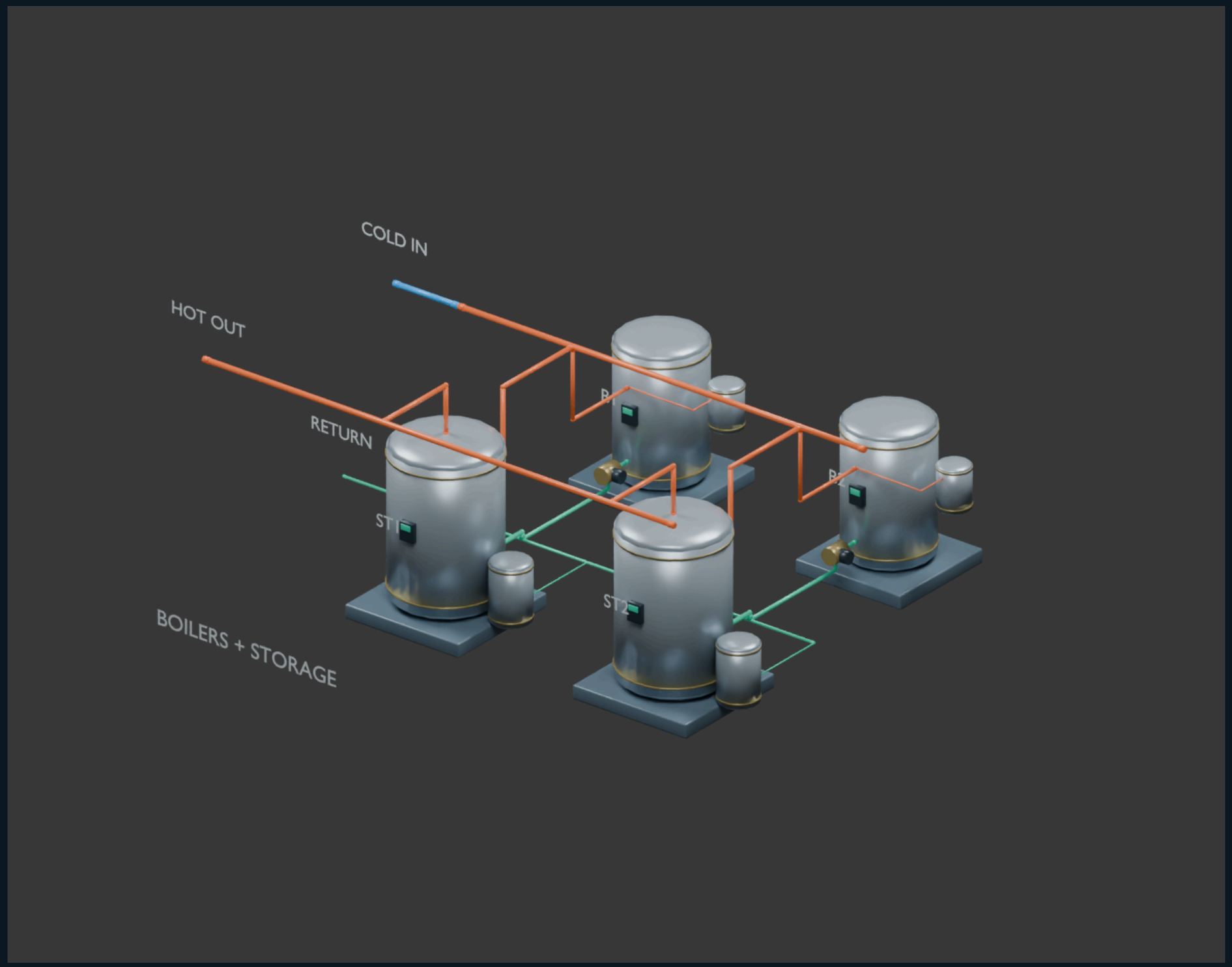
Floor levels now follow A3.11: P2 332.00, P1 342.00, R1 353.00, R2 363.25, R3 373.50, R4 383.75, R5 394.25 ft. The upper guide is PLATE 404.42, not a flat roof. Stack XY and pipe offsets remain assumed.



MODEL STUDY | Unit XY provisionally calibrated; pipe heights and envelopes assumed. Not an as-built.

Plant connections

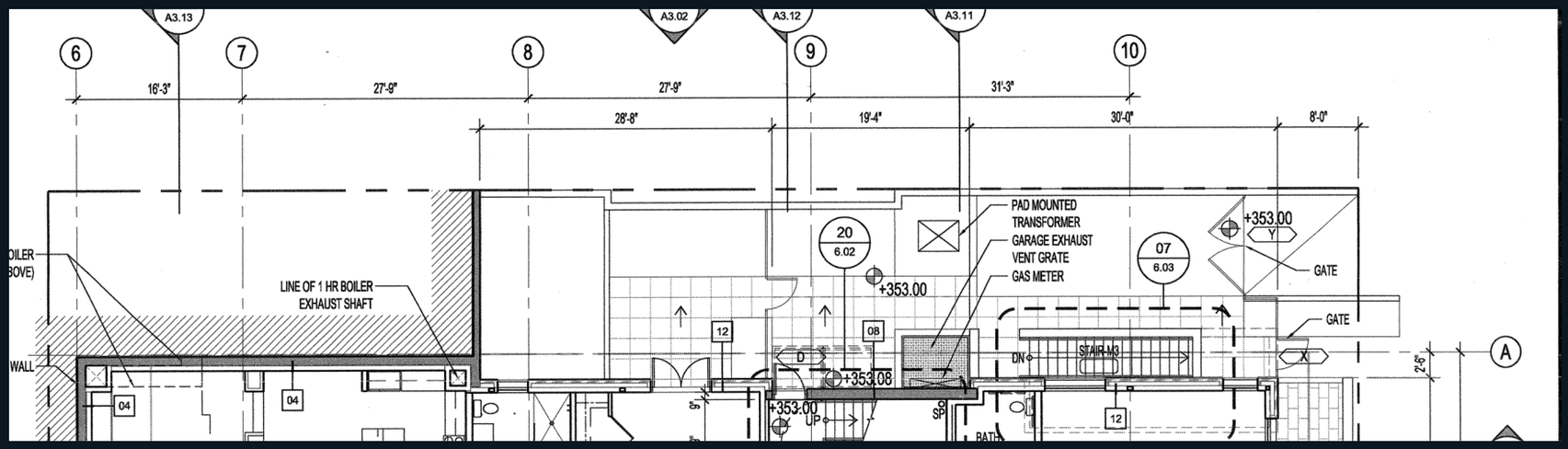
P3.02 / detail 1: two boilers, two storage tanks, four expansion vessels and three installed pumps. WAC2 is a loose spare. Equipment envelopes are representative; relief piping, flues, detailed controls and balancing bypasses are incomplete.



MODEL STUDY | Unit XY provisionally calibrated; pipe heights and envelopes assumed. Not an as-built.

Calibration evidence

A2.03 dimensions the shared structural grid. Those dimensions calibrate P2.02; matched architectural features then transfer scale to the enlarged P4.01 Unit A drawing. The excerpt does not include the P1 architectural plan.

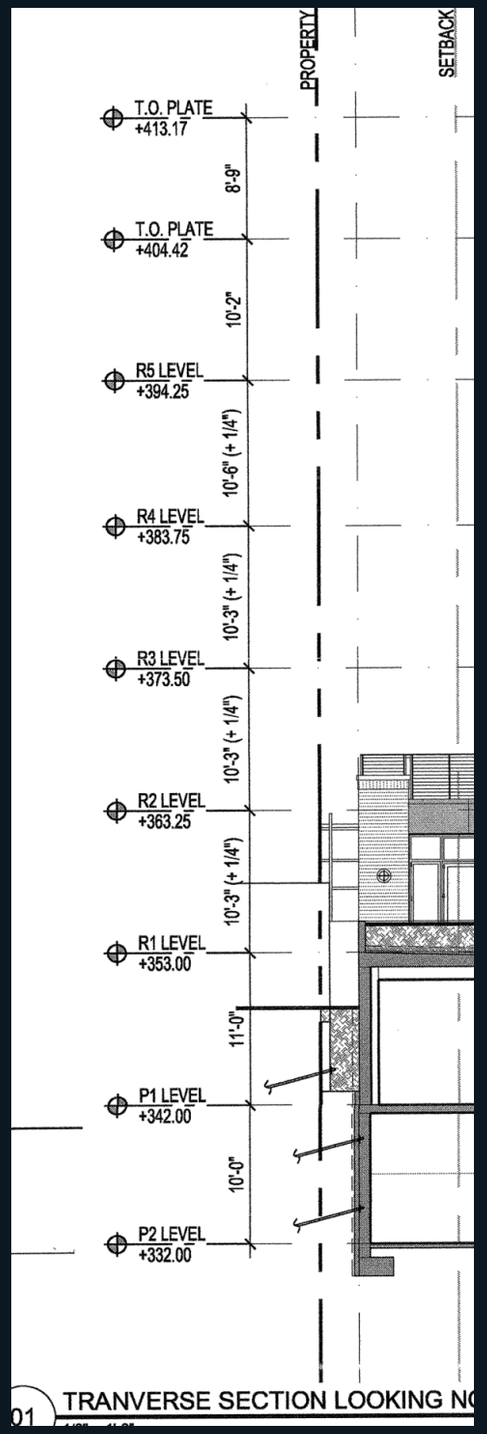


HORIZONTAL GRID
6-7: 16 ft 3 in | 7-8: 27 ft 9 in
8-9: 27 ft 9 in | 9-10: 31 ft 3 in
Overall: 1,236 in across 927 PDF points on P2.02
Scale: 0.75 PDF points / inch

INDEPENDENT VERTICAL CHECK
E-F: 18 ft | F-G: 24 ft 6 in | G-H: 15 ft
All individual spans agree with the working scale within 0.6%.

CROP REGISTRATION
21 matching features; median residual 0.224 PDF points.
Unit conversion: 0.51398 in per normalized pixel.
Plan dimensions increase 28.5% from the earlier placeholder.

DATUM CAUTION
Use the labeled elevations; some interval notes also say (+1/4 in). That small discrepancy is retained for review.
11 ft at Unit 107 is floor-to-floor, not clear ceiling height.



What is checked, and what is still needed

The new construction plans establish building levels and support a provisional horizontal calibration. Concealed pipe placement and equipment geometry still require further evidence.

CHECKED

- Unique IDs and provenance on 319 elements; distribution graphs remain connected.
- Five stack groups, 23 explicit branch groups, 14 Unit A fixtures.
- Seven floor datums match A3.11; Unit 107 floor-to-floor height is 132 in.
- Horizontal and vertical structural-grid spans agree within the raster allowance.
- Cross-sheet registration and model-projected drawing overlay reviewed.

STILL NEEDED

- Unit 107 / P1 architectural and reflected-ceiling plans (not in this excerpt).
- Clear ceiling heights, concealed pipe elevations, offsets and actual shaft positions.
- Equipment dimensions; resolution of the 04 stack R1 and dishwasher connections.
- Clarification of quarter-inch interval notes versus the labeled floor elevations.
- Field records or measurements if installed conditions are the objective.

CURRENT SCOPE

Domestic water supply and return only; sanitary, vent and gas are not modeled.
Working allowances are not survey accuracy: +/-1% scale; +/-6 in local route trace.
No construction lengths or quantities should be taken from this model.